

DECLARATION: I understand that this is an **individual** assessment, and that collaboration is not permitted. I have read and I understand the plagiarism provisions in the General Regulations of the University Calendar for the current year, found at <http://www.tcd.ie/calendar>. I understand that by returning this declaration with my work, I am agreeing with the above statement.

This visualisation was drawn in the Java-based graphical programming environment, Processing, with the aim of recreating Minard's infographic depicting Napoleon's advance to and retreat from Moscow in 1812. Data from the provided *minard-data.csv* file was parsed using Processing's built-in Table class and rendered through a series of layered plots showing troop movement, temperature, and geographic references. The render window is divided into two sections: a troop movement map and a temperature graph.

Troop Movement Map (Upper Graph)

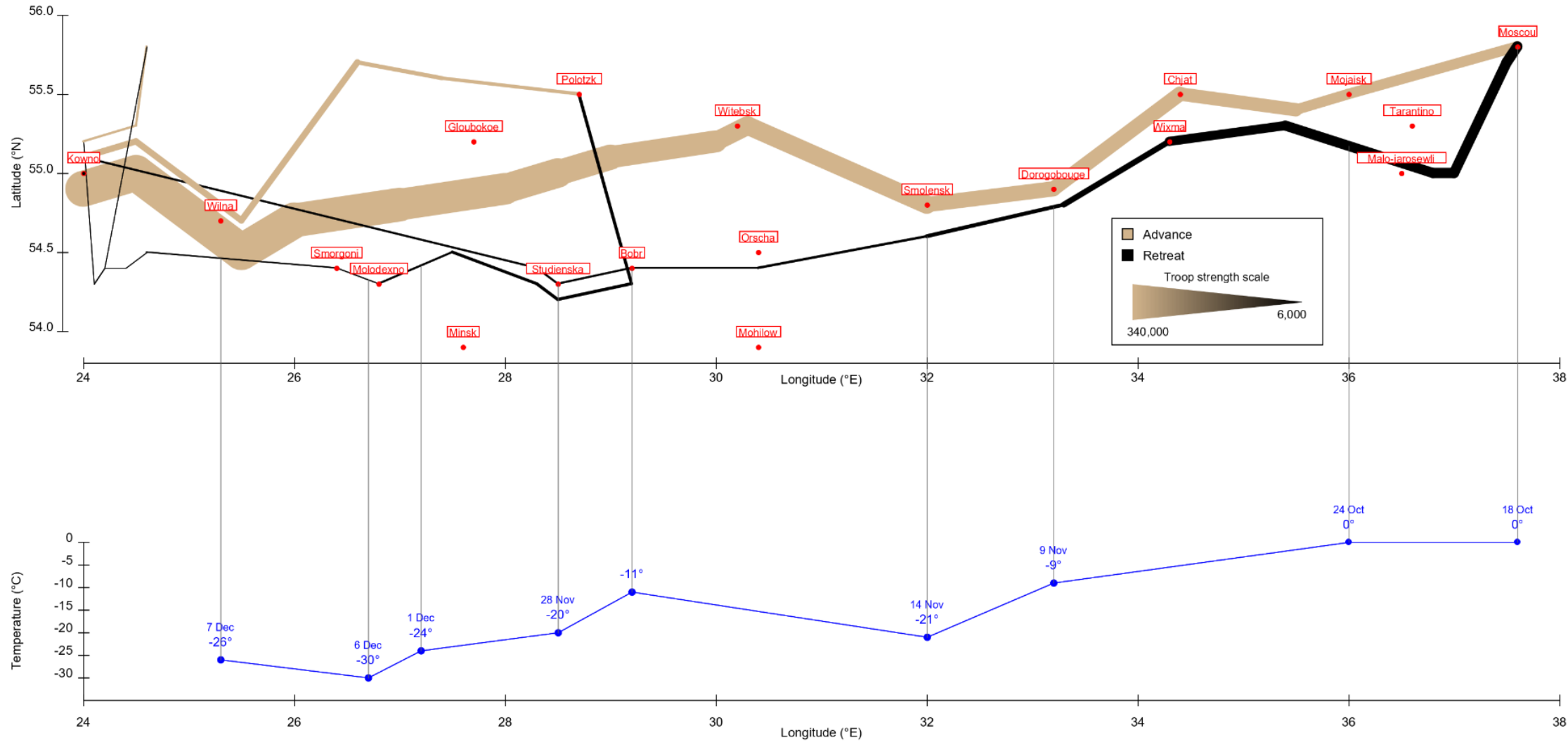
Longitude and latitude coordinates were scaled to the display space using Processing's map() function. Each line segment represents a portion of the army's path while advancing and retreating, with the line colour and thickness representing both the direction of movement (tan for advance, black for retreat) and the number of surviving troops throughout their expedition, respectively. Cities along the route are plotted as small circles with red-labelled text boxes for improved readability, while axes labelled in degrees of longitude (°E) and latitude (°N) provide geographic context.

A legend was added to further clarify the troop movement map. It includes the colours of the lines and their corresponding meanings for the advance (tan) and retreat (black) paths, as well as a horizontal triangular scale representing troop strength. Together with the varying line widths used across the map, this triangle provides a visual representation of the army's decline. The triangle features a smooth gradient transitioning from tan to black, symbolising the reduction in troop numbers from 340000 at the campaign's start to 6000 upon retreat. This was added by interpolating RGB colour values across multiple thin rectangles to simulate a continuous gradient fill.

Temperature Graph (Lower Graph)

This graph plots the relationship between the army's position and the extreme cold encountered during their retreat. The x-axis represents longitude, while the y-axis shows temperature in degrees Celsius. Blue datapoints, connected by lines, form a temperature curve that visually conveys how dropping temperatures coincided with the army's losses. Each data point is annotated with the corresponding temperature and date (day and month) above the marker. To create more of a visual connection between the environmental conditions and the army's physical progression, grey lines were added to link each temperature point to its corresponding geographical position on the retreat path in the troop map.

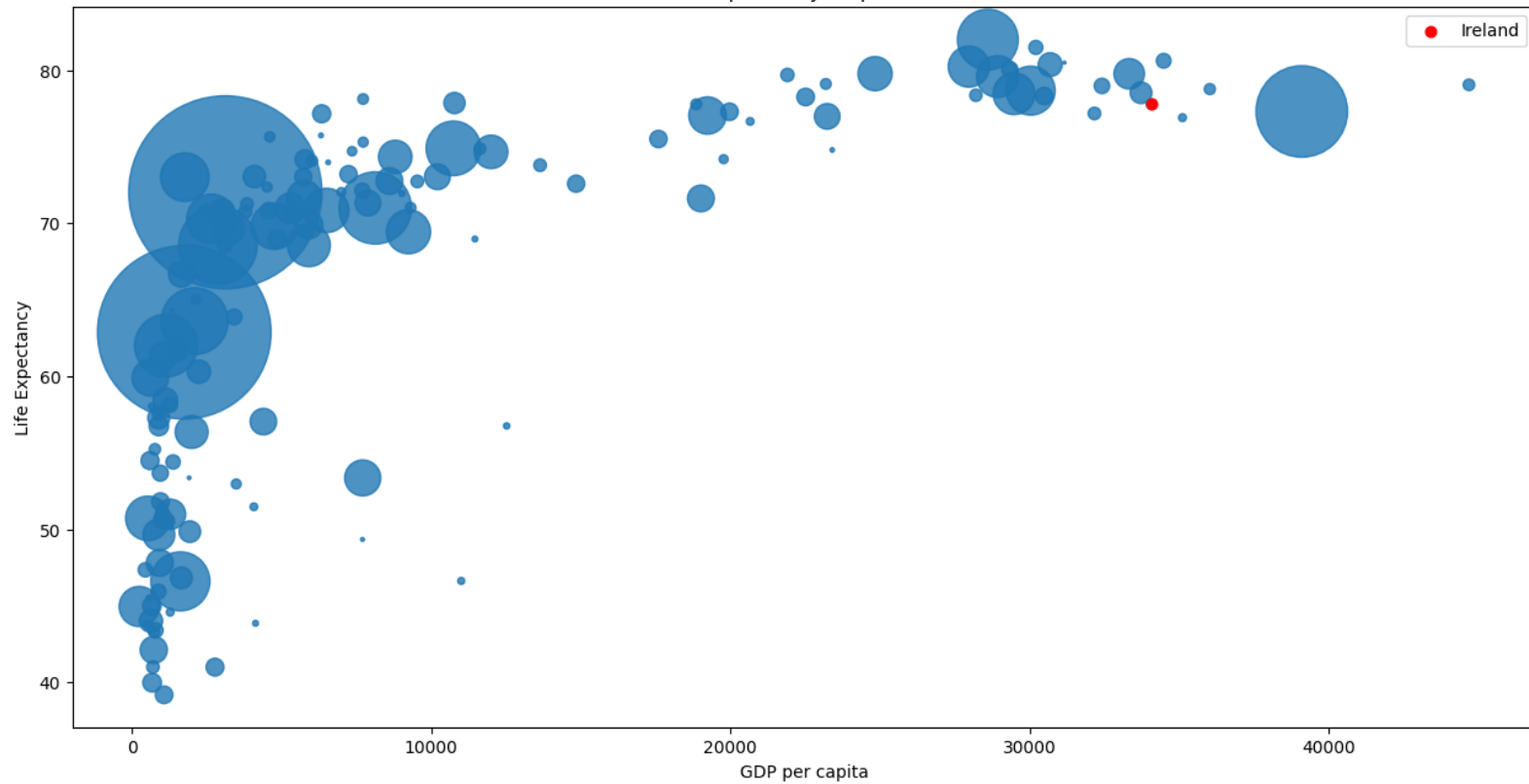
Minard's Map of Napoleon's Russian Campaign (1812)



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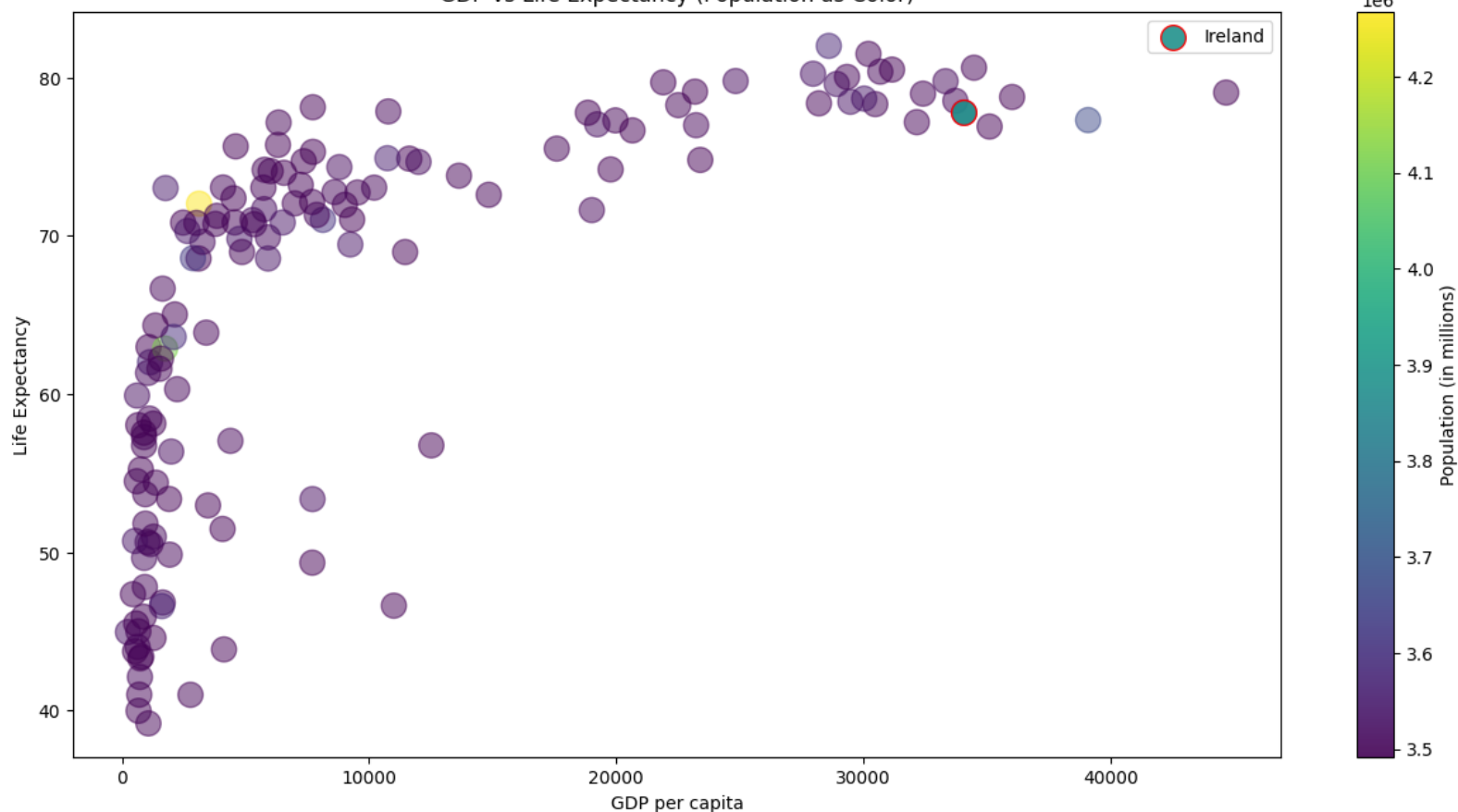
Part A) Chart 1:

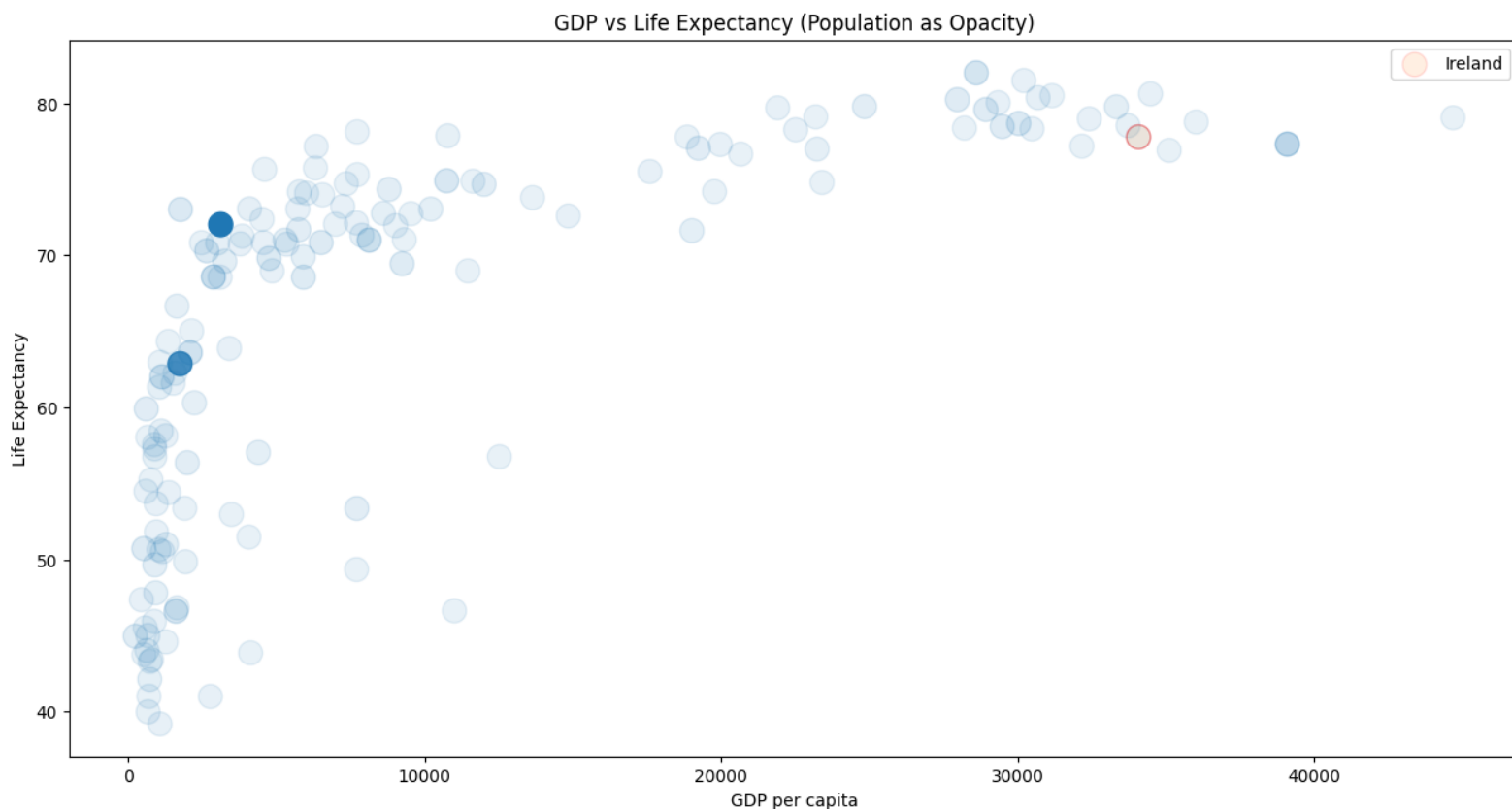
GDP vs Life Expectancy (Population as Size)



Part A) Chart 2:

GDP vs Life Expectancy (Population as Color)



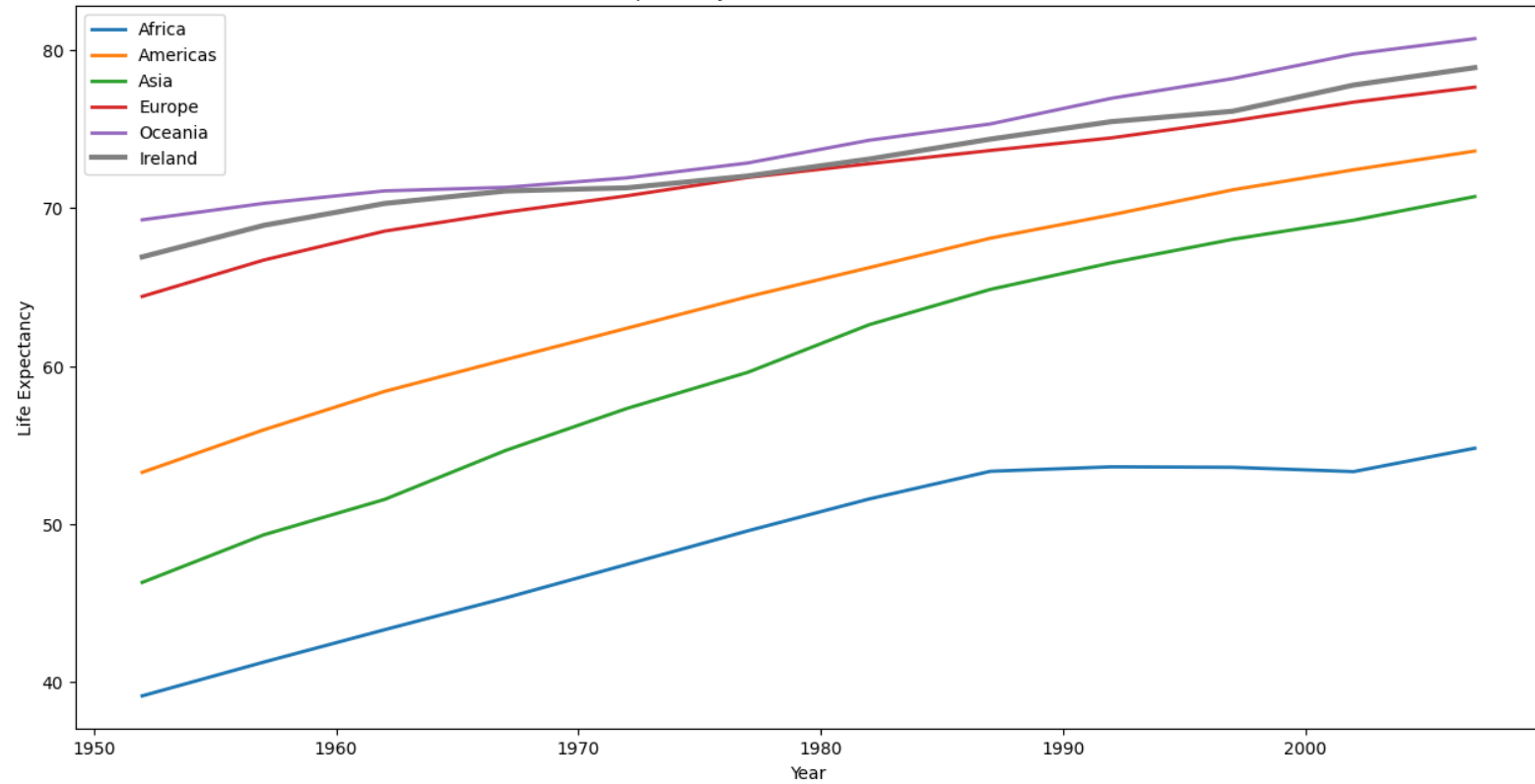
Part A) Chart 3:

These visualisations were produced in Python using Pandas for data handling and Matplotlib for plotting. As per the instructions, the dataset was filtered to include only data from 2002, allowing the relationship between GDP per capita and life expectancy to be examined across countries. Together, the charts reveal a correlation between national wealth and health outcomes, with wealthier countries tending to have longer life expectancies.

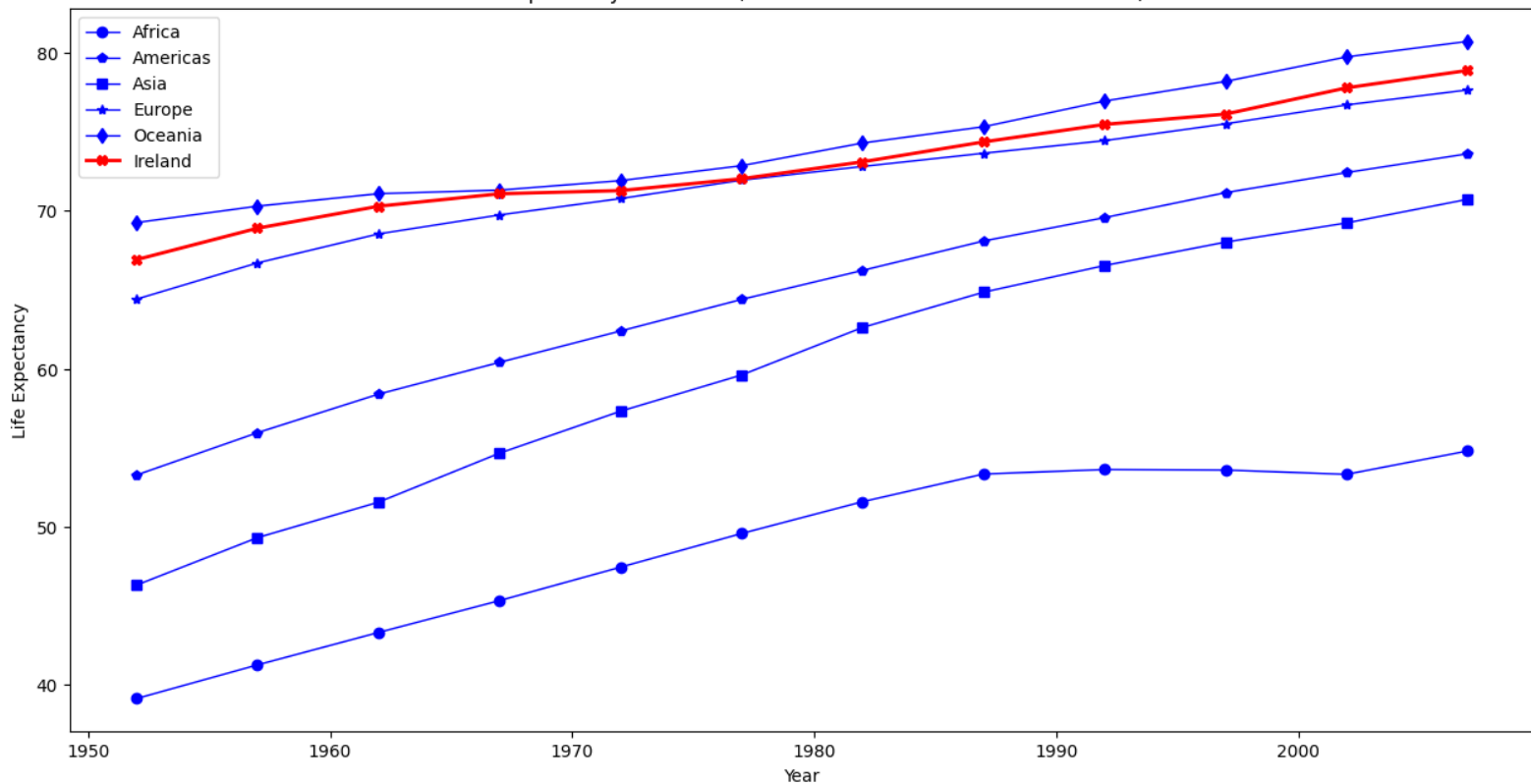
Population is encoded in three different ways across the charts: by size (Chart 1), colour (Chart 2), and opacity (Chart 3). Using size gives an easy to grasp sense of relative magnitude but can cause densely populated countries to dominate smaller ones visually. Colour mapping resolves most overlap, since all markers share equal size, but makes precise population differences harder to interpret. Opacity highlights dense clusters effectively. While it makes identifying densely populated countries easier, it offers more of a qualitative impression of population concentration rather than a precise quantitative one. Ireland is distinctly marked in each case with a red outline, while other countries remain anonymous for clarity.

Part B) Chart 4:

Life Expectancy over Time (Continent as Color)

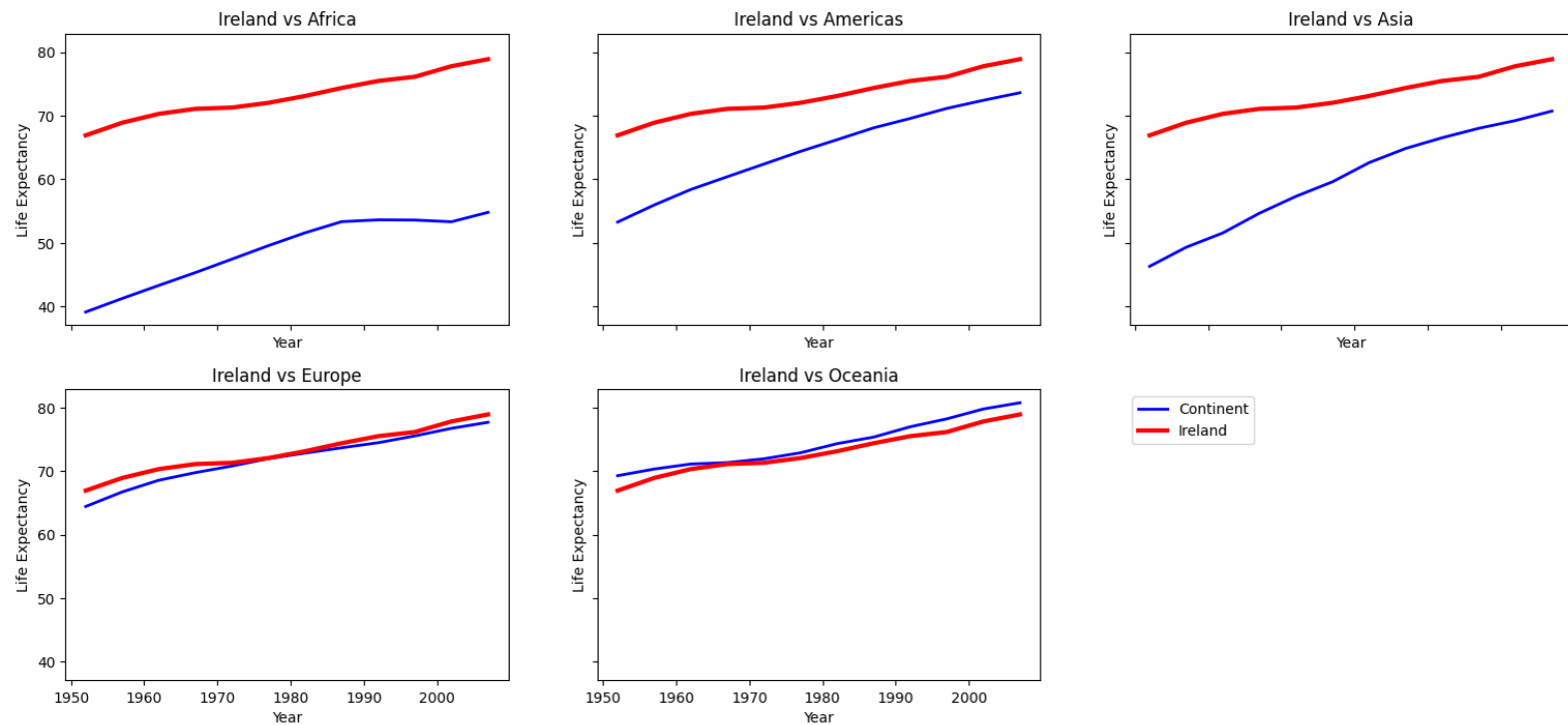
**Part B) Chart 5:**

Life Expectancy over Time (Continent as Line-Marker Combination)



Part B) Chart 6:

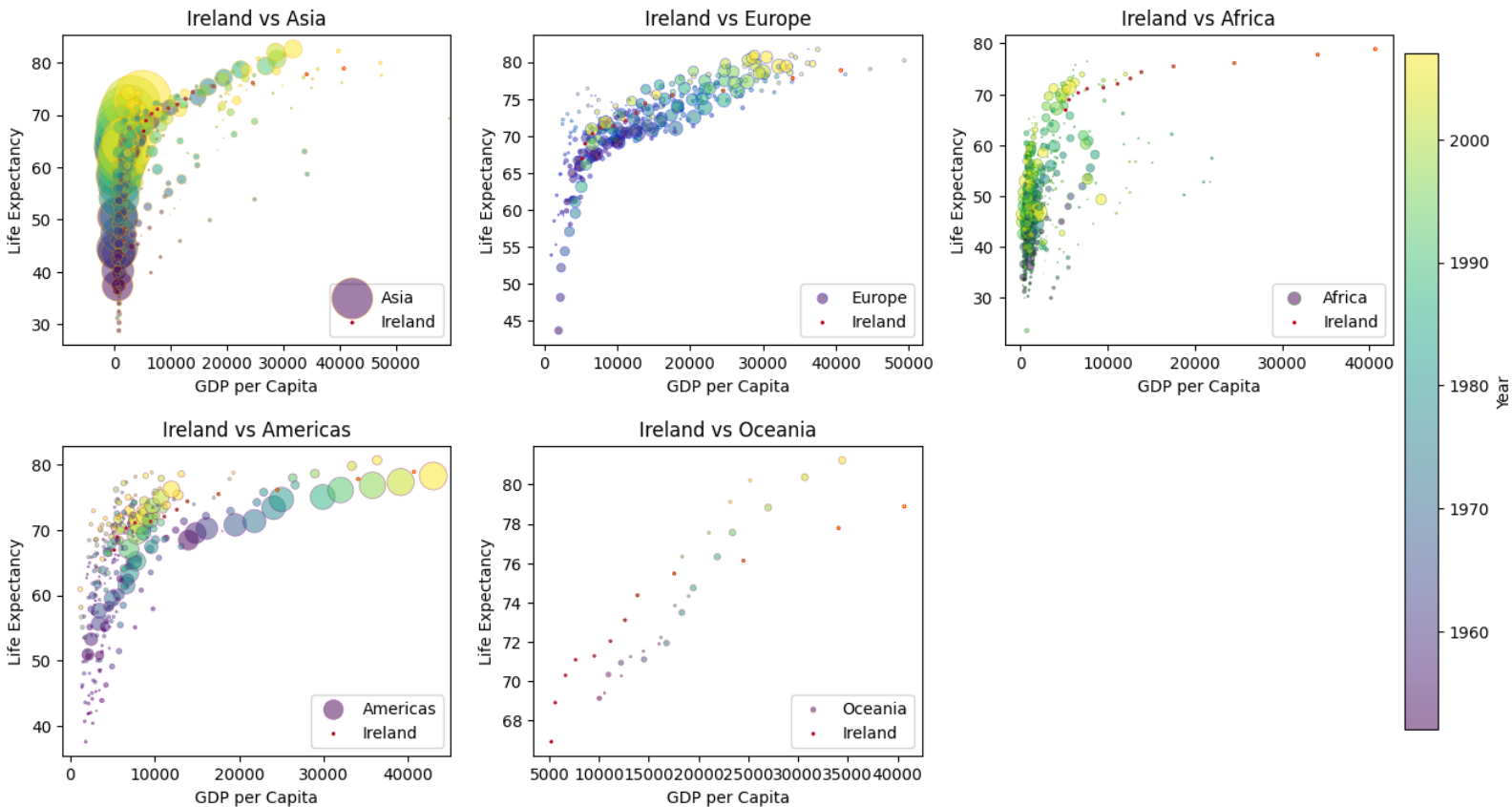
Life Expectancy over Time (Continent as Facets)



These charts use aggregated life expectancy values by continent and year (1952–2007) to illustrate how the life expectancy of each continent, and by extension, global health, has evolved over time. Matplotlib was again used to visualise the data through three different encodings: colour, marker style, and faceting, while keeping year and life expectancy consistently mapped to the same axes.

Colour (Chart 4) allows for quick visual comparison across continents but becomes less effective in monochrome contexts (e.g., black-and-white print) or for viewers with colour-vision impairments. Marker styles (Chart 5) perform better in such contexts and provide clear shape-based distinctions, though they can appear cluttered when many points overlap. Faceted plots (Chart 6) simplify the visualisation by displaying each continent in a separate subplot, preventing overlap and revealing clearer within-continent trends, albeit at the cost of direct cross-continent comparison.

Due to the colour-based encoding in Chart 4, Ireland is intentionally highlighted in grey to contrast with the other coloured continent lines, whereas it is highlighted in red for Charts 5 and 6. As there are fewer continents to convey than there are countries in Charts 1-3, each continent is labelled in the legend and subplot titles for improved interpretability.

Part C Chart 7:**Chart 7: Exploratory Visualization - Ireland vs Continents (1952-2007)**

The final chart combines all five attributes of the dataset, continent, year, life expectancy, population, and GDP per capita, into a multifaceted exploratory dashboard. Initially, this visualisation was developed as a single scatter plot displaying all continents together; however, the result appeared visually cluttered and made it difficult to distinguish Ireland. To address this, the design was reformed into a faceted layout, akin to Chart 6, providing separate panels for each continent.

Within each panel, GDP per capita is mapped to the x-axis, life expectancy to the y-axis, population size to marker size, year to a continuous colour gradient, and continent to both the edge colour and subplot title. By using all these encoding methods at once, the relationship between economic growth, population size, and health across time has been captured. To keep Ireland visible without being overwhelmed by each continent's data points, Ireland's data is highlighted with a red outline in each facet.